Anna Zheng - Reflection 3

For our final project, my partner and I prototyped an automated landslide barrier wall aimed at improving disaster detection and physical defense in vulnerable, mountainous regions like Nepal, which recently experienced severe flash floods and debris flows due to a major landslide. Our goal was to design an early detection system that triggers rapid physical deflection to redirect falling debris away from populated areas. To achieve this, our system incorporated various sensor inputs feeding into an Arduino Nano ESP32 microcontroller. We utilized three distinct inputs: a custom moisture sensor to detect high soil saturation, a photoresistor sensor to detect debris movement, and a manual toggle switch for operator override. For outputs, we connected a servo motor programmed to rotate a wall, constructed from popsicle sticks and duct tape, from 0º to 90º upon deployment, alongside a flashing LED that served as a visual warning alarm.

To complete our prototype within our three lab sessions, my partner and I divided our responsibilities based on our individual strengths while maintaining collaboration. My partner took primary responsibility for assembling the breadboard wiring, managing the circuit configurations, and setting up the voltage divider for our sensors with help from course staff. Meanwhile, I focused on designing the physical barrier wall structure, mounting it to the servo motor arm, and took charge of writing our code with AI. Coming into this project, I found working with microcontrollers to be quite intimidating. However, actively watching my partner wire the components and receiving guidance from course staff made the hardware integration process much more approachable and helped me gain a solid understanding of how microcontrollers interact with physical circuits. After every lab session, we committed our changes onto our GitHub repository.

A highlight of my experience was utilizing generative AI to write and debug our MicroPython software in the Arduino Lab IDE. I used AI prompts to program the various components of our prototype, such as the servo arm's 0º to 90º rotation and the repeating LED flash sequence. AI was also instrumental in brainstorming mechanical configurations for mounting the physical wall securely to the servo arm. Through this hands-on coding process, I learned how to use AI tools more responsibly and efficiently. Initially, vague prompts resulted in incorrect code outputs. I quickly realized that receiving functional code required precise prompts that defined the microcontroller board, the MicroPython language, the assigned GPIO pin numbers, and the conditional logic of each component.

We encountered several challenges during testing. Mechanically speaking, attaching the wall to the servo arm so that it opened cleanly when triggered took significant trial and error alongside AI assisted ideas. Furthermore, our custom moisture sensor was extremely finicky across the three lab sessions due to fluctuations in room humidity and general sensor

unreliability. We had to continuously adjust our threshold value, evolving our measurement method from raw ADC values to voltage conversions, before ultimately settling on measuring resistance in ohms through a voltage divider circuit. Once we calibrated the threshold to trigger at 50 ohms, the system was decently reliable. In our final demonstration, the prototype successfully met all requirements by rapidly deploying the deflection wall and activating visual warnings using both automated and manual triggers. Seeing the prototype actually work was incredibly rewarding because going into this module, I never thought building a functional embedded system was something I could accomplish. In the future, this concept could be scaled up using heavy-duty materials and an audible siren to give communities time to evacuate during natural disasters.